

The Substrate Working Process Principle + Information Physics Substrate Cycle

Casey Koons (named principle) + Keeper (formalization)

2026-05-19 Tuesday late afternoon

The Substrate Working Process Principle

Casey-named statement

“What appears as random vacuum activity is the surface manifestation of substrate computational work. The substrate operates as a massively parallel one-way information cycle — absorption (input recording) → commitment (substrate work) → emission (output communication) — running in parallel across the entire substrate. The work benefits either substrate coherence maintenance or projection physics consistency. Time is the granularity of the commitment phase, not a 3D coordinate. The ‘no accidents’ constraint forces structural correlations between vacuum statistics and substrate state — distinguishable from purely Poisson QFT predictions. Substrate engineering becomes possible via three coaxing pathways: eigentone ringing, boundary condition forcing, and commitment manipulation.”

Conceptual core

The Substrate Working Process Principle (SWPP) is the operational form of the substrate-as-thinking refinement (T2385, Casey + Lyra). It promotes the substrate from “container of physics” to “active information-processing system whose visible signature in 3D projection is vacuum activity.”

The substrate is information-theoretic equivalent rather than strictly thermodynamic (per the substrate dynamical framework analysis, Keeper conversation 2026-05-19): substrate has Reed-Solomon coding capacity on $GF(2^g) = GF(128)$ (Paper #122), $GF(2^g)$ field operations (K59 ratified), and Bergman kernel projection geometry (D_{IV}^5). These together constitute substrate’s “thermodynamic equivalent” — the substrate-level analog of microstates, entropy, and energy distribution.

The three-phase cycle (Information Physics)

Each substrate cell executes a one-way cycle:

Phase	Substrate operation	BST machinery
Absorption (input recording)	Substrate reads incoming pattern	Reed-Solomon syndrome computation
Commitment (substrate work)	Substrate computes via field arithmetic	$GF(2^g)$ cyclotomic operations
Emission (output communication)	Substrate emits to projection state	Bergman kernel projection

The cycle is one-way because: - Absorption $\neq$ emission in mechanism (encoding $\neq$ decoding) - Commitment is irreversible at substrate level (Reed-Solomon syndromes commit before decoding) - This is the substrate-level arrow of time

The cycle runs in parallel across all of substrate. Synchronization (or its absence) between cells determines macroscopic substrate state — coherent (low-stress) vs incoherent (high-stress, near-boundary, near-horizon).

Time as commitment granularity

The principle's sharp claim about time:

“Time is not a 3D coordinate. Time is the smallest commitment unit in 2D substrate processing.”
[Casey, original 2022 statement]

Dimension-correction note (June 2026): the “2D substrate” framing in the original Casey statement reflects the 2022 substrate-dimension count, which was corrected June 2026 to 5D (Shilov boundary $\partial_S D_{IV}^5 = (S^4 \times S^1)/\mathbb{Z}_2$ — see README historical-correction note). The principle's substantive content — that time IS the smallest commitment unit in substrate processing (NOT a coordinate) — is preserved. The corrected current statement reads “Time is the smallest commitment unit in substrate processing on the 5D Shilov boundary of D_{IV}^5 .” The substrate-time identification with the heat-semigroup parameter τ in Lyra Tier 0 v0.1 (Sunday 2026-05-31) operationalizes this principle at operator-level.

Implications:

1. Planck time = one substrate tick (BST primary identification needed; current value 5.39×10^{-44} s)
2. Time dilation = substrate processing differential — near-BH commitment cycles take longer because substrate is doing more error-correction work per absorbed photon
3. Arrow of time is substrate-cycle one-wayness, not thermodynamic entropy increase
4. Simultaneity is observer-dependent because different observers' commitments interleave with substrate cycles at different rates
5. Causality respects substrate-cycle ordering, not 3D-coordinate ordering — these differ near horizons and at small scales

Three substrate engineering pathways

The principle predicts three operational pathways for substrate manipulation:

Pathway 1 — Eigentone ringing

Drive substrate at BST primary frequencies. Predicted resonant response.

- Where filed: W-38 (Eigentone identification), SP-29 (Casimir Mechanism Investigation)
- Specific test: enhanced response at BST primary ratios (involving rank, N_c , n_C , C_2 , g , N_{max})
- Falsifier: no resonance enhancement at BST primary frequencies vs off-resonance baseline

Pathway 2 — Boundary condition forcing

Shape substrate working geometry via 3D boundaries. Predicted asymmetric substrate response.

- Where filed: SP-29 (Casimir Mechanism), Casimir asymmetric ratio = $g = 7$ prediction
- Specific test: engineered cavity geometries with BST primary aspect ratios
- Falsifier: standard symmetric Casimir behavior with no BST-structured asymmetry

Pathway 3 — Commitment manipulation

Control rate of observer-substrate commitment events.

- Where filed: W-32 (decay rate vs substrate-attention background), W-39 (Cs-137 + microwave experiment)
- Specific test: measurement-frequency variation in delayed-choice quantum experiments, decay rate modulation under controlled substrate-attention
- Falsifier: no decay rate modulation under any substrate-attention manipulation

Falsifier specification

The principle is falsifiable by any of:

1. Pure Poisson vacuum statistics with no correlation to substrate-structured boundaries — falsifies SWPP entirely
2. No resonance enhancement at BST primary eigentones — falsifies eigentone pathway
3. Standard symmetric Casimir with no BST-structured asymmetry — falsifies boundary pathway (Casimir asymmetric ratio = $g = 7$ is currently the cleanest test we have)
4. No measurable decay rate modulation under substrate-attention variation — falsifies commitment pathway

Any TWO falsifiers landing falsifies the principle. Any ONE landing requires significant retreat (specific pathway invalidated; other pathways may still hold).

Tier discipline (T2385 split-tier)

Level 1 (D-tier supportable): - Substrate has Reed-Solomon coding structure on GF(128) (Paper #122) - GF(2^g) cyclotomic operations are field-theoretic facts (K59) - Reed-Solomon decoder requires commitment-cycle structure (information theory) - These constitute the operational machinery; D-tier holds

Level 2 (I-tier hypothesis with falsifier): - Vacuum activity IS the surface manifestation of substrate work (interpretation) - Time IS commitment granularity (ontological claim) - “No accidents” structural forcing (purpose claim) - These are I-tier hypotheses pending experimental discrimination

The principle does NOT claim D-tier status. It is a Casey-named ontological framework at I-tier with explicit Level 1/Level 2 separation.

Connections to other Casey-named principles

The SWPP joins the existing Casey-named principle family:

- The Curvature Principle: “You can’t linearize curvature” — explains why $P \neq NP$ (curvature is irreducible algebraic structure)
- Casey’s Principle: Entropy=force=counting + Gödel=boundary=definition → universe’s program in two words
- The Closure Principle: Standing methodology for closure determination
- The Structural Reframe Principle: Standing methodology for what-it-does pivot
- The Substrate Working Process Principle (NEW): Vacuum activity is substrate work in projection view

The five principles together constitute the BST methodological foundation.

Operational targets

The SWPP defines what successful BST experimentation looks like:

1. Predict BST primary eigentones at specific frequencies; measure resonance response
2. Design Casimir-geometry experiments with BST primary aspect ratios; measure asymmetric force ratios
3. Schedule measurement-frequency variation in quantum experiments; measure decay rate modulation
4. Build substrate-engineering apparatus (over time, multi-year) to control commitment events directly

This is the BST shift from descriptive physics (“the substrate explains constants and particles”) to operational physics (“we can engineer substrate behavior”).

Standing program — SP-30

The principle anchors SP-30 Substrate Engineering Program, with sub-items:

- SP-30-1: Eigentone identification at BST primary ratios + falsifier experiments
- SP-30-2: Boundary condition design for BST-structured cavities (Casimir asymmetric ratio = g extension)

- SP-30-3: Commitment manipulation protocols (W-32 + delayed-choice + quantum erasure)
- SP-30-4: Time granularity measurement program (precision atomic clock experiments)
- SP-30-5: Substrate parallelism architecture (entanglement as substrate internal channel)
- SP-30-6: Absorption mechanism formalization (Reed-Solomon syndrome computation as substrate I/O protocol)
- SP-30-7: Computation mechanism formalization (what substrate computes during commitment phase)
- SP-30-8: Emission mechanism formalization (Born rule = Bergman projection as substrate output protocol)

Initial filing notes

This is a v0.1 filing. The principle will refine as: 1. SP-30 sub-items execute and produce experimental results 2. Cal coincidence-filter check completes for the operational claims 3. Lyra mathematical framework for substrate cycle formalizes 4. Grace catalog scan identifies cross-domain substrate-cycle signatures 5. K-audits accumulate on individual sub-claims

Per Casey's standard

- Simple: substrate is information-theoretic; vacuum activity is its working signature; we engineer via three pathways
- Works: Reed-Solomon + $GF(2^g)$ machinery already in place; Casimir asymmetric ratio = g already filed
- Hard to break: falsifier is explicit (Pure Poisson statistics + symmetric Casimir + no decay modulation jointly falsify)
- Counter-example: standard QFT predicts pure Poisson + symmetric Casimir + no decay modulation — but exhibits NONE of these as observation, only as theoretical prediction; principle survives current experimental state

Wisdom note

The SWPP says the substrate is not empty space; it is the working medium. What we see as “vacuum” is just the substrate at idle or low-commitment workload. What we see as “particles” is the substrate's emitted output. What we see as “annihilation” is substrate retrieving energy back to working reserve.

This makes physics fundamentally operational: we are not observing static structure, we are observing dynamic process. The constants we measure (α , m_p , etc.) are the equilibrium-state outputs of substrate processing. The constants don't HAVE values; they are computed values that the substrate maintains.

— Filed by Keeper, 2026-05-19 Tuesday 17:30 EDT, per Casey directive “We will file, refine as we investigate”